School K-8 Prototype Scorecard Spreadsheet Review

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Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Zone volumes calculations are wrong (K8S27)
- Some exterior wall U-values incorrect (K8S41)
- One SHGC value incorrect (K8S44)
- Some heat pump EERs and heating COPs incorrect (K857, K858)
- Refrigerated spaces should not have ventilation (K8S69)
- Restroom ventilation based on exhaust not cfm/person (K8S70)
- Kitchen exhaust ventilation missing (K8S71)
- Number of parking spaces per building area incorrect (K8S88)
- Number of elevators and power multiplier inconsistent (K8S95)

The following are the highest priority comments:

- A 7% window-to-wall ratio is very low (K8S18, K8S43)
- About three times the number of occupants as expected with many zones using twice the occupancy density as required (K8S33, K8S36, K8S68, K8S76)
- Wrong system type for large multipurpose room (K8S55)
- Static pressure is too high for rooftop units (K8S62)
- Outdoor ventilation of 30 cfm/person instead of 15 cfm/person was used (K8S67)

The following comments represent other important issues found:

- Second floor outdoor walkways missing for shading (K8S16)
- The number of skylights is very low (K8S29)
- Separate air handlers for each zone are unlikely (K8S30)
- Concrete or concrete block is typically used in rooms serving as gyms (K839)
- Infiltration should be calculated for each building separately (K8S52)
- Non-operable windows in classrooms are unusual (K8S53)

- Roof replacement frequency of 30 years is too optimistic for budget constraints in schools (K8S54)
- Interior lighting replacement frequency of 10 years is too optimistic for budget constraints in schools (K8S84)
- Four entrances for exterior lighting calculation too low for four buildings with exterior corridors and doors from each classroom (K8S87)
- No student pickup exterior lighting power allowance included (K8S91)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- How was the floor area, geometry, number of floors, WWR, and floor height determined? (KS3S, K8S17, K8S20, K8S22, K8S51, K8S14, K8S15)
- How were the vintages of the existing schools decided (K8S6)
- Why is the school K-8 not a PK-5 or middle school? (K8S7)
- How many students and how many classrooms? (K8S21)
- Where are the various zones located? (K8S26)
- Are economizers being used? (K8S63)
- How were the classroom thermostat setpoints selected? (K8S66)
- How were the number of elevators chosen (K8S94)

Background

The review of the "scorecard," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Interior Lighting
- Equipment
- Ext Lighting
- Schedules

- Energy Usage
- Weights

The comments on these tabs are labeled using the following format **K8Sx-**<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after the Schedules tab.

A copy of the spreadsheet is available that includes which cells on each tab were checked against the sources and were consistent (shown in light green) and which had anomalies and are described in a comment below (shown in tan).

K8S1-Notes (C6)

The link for "CEC School prototype (CBECC 2022.3.1)" is shown as:

https://bees.noresco.com/software2022.html

but it gets redirected to:

https://www.energy.ca.gov/programs-and-topics/programs/building-energy-efficiency-standards/2022-building-energy-efficiency-1

which is a generic CEC page about 2022 Energy Code Compliance Software.

K8S2-Notes (C19)

The link for "Air leakage Memo, NORESCO, 2025," which is shown as:

Infiltration_rate_report_14May2025-Short-Version.docx

does not seem to work (for me at least).

K8S3-Notes (n/a)

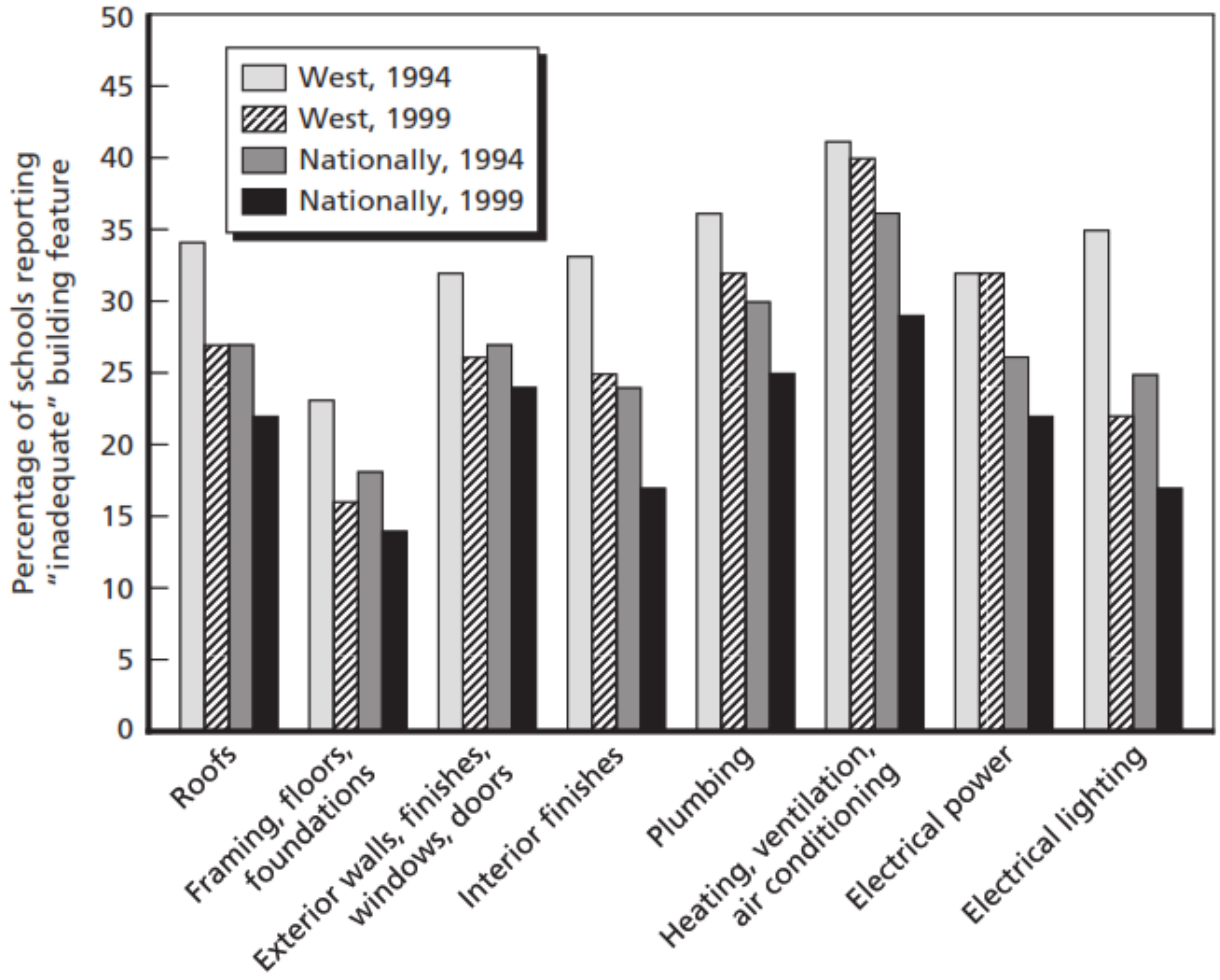
Since the geometry changed has such a different basis from the older PNNL prototype, it would probably be good to expand on how that was derived and link to any documentation on it.

K8S4-Notes (n/a)

The capital constraints on some school districts, especially in states like California where voters have such a strong say on funding, often result in using older buildings well after they should be replaced, often with poor maintenance histories. This reality will mean that it is very likely that older equipment (lighting, HVAC, etc.) is being used much longer than typically expected for schools. The following graph from Rand titled "California's K-12 Public Schools How Are They Doing" (2005) is one of several substantiating the lack of investment in California. While the report is a little dated, it is possible that these problems have actually gotten worse.

https://www.rand.org/pubs/monographs/MG186.html

Figure 5.1
Percentage of K–12 Public Schools Reporting "Inadequate" Building Features, the West and the United States, 1994–1995 and 1999–2000



SOURCE: U.S. General Accounting Office, 1995a; and U.S. Department of Education, NCES, 2000.

K8S5-Notes (n/a)

To provide a thorough review of the prototype, I reviewed several great resources shown below prior to beginning my review. Originally, 80 resources were identified, but the following seemed the most important.

Sources that *should* be reviewed for background information and are fundamental for reviewing the school prototype are the following:

- ASHRAE Handbook HVAC Applications – Chapter 8 Educational Facilities (2023)
- ASHRAE Research Project Report 1814-RP Actual Energy Performance of Secondary Schools Designed to Comply with ASHRAE Standard 90.1-2010 (2023) https://store.accuristech.com/standards/rp-1814-actual-energy-performance-of-secondary-schools-designed-to-comply-with-ashrae-standard-90-1-2010?product_id=2524559
- Development of the Advanced Energy Design Guide for K-12 Schools – 50% Energy Savings (2013) https://docs.nlr.gov/docs/fy13osti/51437.pdf
- Technical Feasibility Study for Zero Energy K-12 Schools (2016) https://docs.nlr.gov/docs/fy17osti/67233.pdf
- California K-12 and Community College Zero Net Energy Retrofit Readiness Study (2017) https://www.calmac.org/publications/Nbi171030CPUCk14.pdf
- Building Type Basics for Elementary and Secondary Schools (2001)

Additional useful sources could be added to the list shown in the scorecard spreadsheet:

- Final Evaluation Report for The Alliance to Save Energy Green Schools Green Communities 2002-2003 School Programs (2004) https://www.calmac.org/publications/02-03_ASE_Final_Report_fnl3.pdf
- Culver City Unified School District Facilities Master Plan (2014) https://www.ccusd.org/CapitalProjects/CCUSD_Facilities_Master_Plan.pdf
- Designing for Zero Carbon Volume 3 Case Studies of All-Electric K-12 Schools (2025) https://calbem.ibpsa.us/wp-content/uploads/2025/09/Designing-for-Zero-Carbon-V3_K-12-Schools-2.pdf
- California Schools Market Characterization (2005) https://www.calmac.org/publications/PGE_California_Schools_Market_Characterization.pdf

- Architectural Graphics Standards Twelfth Student Edition (2017)
- Elemenary/Middle Schook PK-8 Typical Space Program https://www.dpi.nc.gov/documents/schoolplanning/space-profile-pk-8/download
- California's K–12 Public Schools How Are They Doing? (2005) https://www.rand.org/content/dam/rand/pubs/monographs/2004/RAND_MG186.pdf
- National Center of Educational Statistics Digest of Education Statistics https://nces.ed.gov/programs/digest/current_tables.asp

K8S6-Notes (A20:F32)

While a graph of the construction rates is provided, no subsequent analysis has been added as to what vintages make the most sense for the historic construction trends.

K8S7-Notes (A20:F32)

Implied by using a K-8, kindergarten through eighth grade, prototype model is that that is the most common type of building for those ages of school children. Please provide some statistical source to back up this assertion. For many school districts, PK-5, prekindergarten to fifth grade, are different buildings than middle schools, typically sixth grade to eighth grade. This matters in terms of the size of the building and the types of spaces in the building and the people density. Younger grades usually have fewer people in a classroom and are likely to have fewer computers. I believe it is becoming even more common for urban school districts to have separate middle schools.

K8S8-Notes (A20:F32)

Some primary and middle schools offer specialization (STEM and art/maker focused are probably the most common), and while these are probably not a good match for a prototype, it would make sense to explicitly state that the prototype is not a specialized school. Perhaps describing it as "conventional curriculum" would be appropriate.

K8S9-Notes (A20:F32)

For most school districts, the taxes collected by residents make up a substantial portion of the school funds for both operating and maintaining existing schools and constructing new

schools. This results in real disparities between schools that could be captured in a prototype. The age of building components, including HVAC, number of students per classroom, specialized classrooms, and physical education facilities, may vary substantially based on the funding of the school. Given that a prototype could capture these differences or try to average across these differences, it should be explicitly stated the assumption that is being used. Perhaps describing this as an averaged school across a diversity of school funding situations might clarify.

K8S10-Notes (B23)

The graph shows "STC: 41, 49" please explain what this means or remove it.

K8S11-Notes (n/a)

Many of the abbreviations being used on the data sources are familiar to building energy analysts but may not be as familiar to others. Please add an unabbreviated version of each item.

K8S12-Notes (A1:F19)

If this list of data sources is intended to be comprehensive, some have been missed (Title 20, Plug Load project, etc.) and some are included here that don't seem to be used such as CEUS and ComStock. It also seems odd that no specific school data sources are being used since there is a variety of data available specifically for schools. Please review all citations and update this list appropriately.

K8S13-Notes (A1:F19)

Please add links for all sources.

K8S14-Notes (n/a)

In general, other scorecards reviewed so far have had substantially more information on the Notes tab. This notes tab seems to not include information about the construction rate over time, justification of area or number of floors, window-to-wall ratio, why representative years were chosen for different vintages, other graphs supporting major assumptions,

alternative data sources to help justify major assumptions, or primary energy source. In general, the notes seem weak compared to other scorecards. This is substantially the same comment as NURS8.

K8S15-Prototype (C7:C16)

For shape, total floor area, aspect ratio, number of floors, roof type, WWR, shading geometry, floor-to-floor height, floor-to-ceiling height, and glazing sill height, the data source is shown as from CEC. From the Notes tab, the only reference to CEC is for the CEC School Prototype, which is shown with CBECC 2022.3.1 in parentheses and the URL that was no longer pointing to the original location (see K8S1-Notes). The use in CBECC is not sufficient to explain why those values were chosen. An analysis deriving the assumptions of the K8 school building by CEC for CBECC should be documented somewhere, and that should be used as the source of the assumptions. I understand that such a document is under development, so this comment may not be relevant in the future. The justification for those individual assumptions should be reviewed when ready.

K8S16-Prototype (A13:C13)

Some shading should be added for the courtyard side of the two-floor buildings, north classroom and west classroom to represent an outdoor walkway. The second floor shows doors to the courtyard side of the building, plus no interior corridors are shown on the Zones plan diagram. Given this, it is likely that exterior corridors that would shade the first floor should be present and are missing. It is also very common in this configuration to have shading over the second-floor corridor as well.

K8S17-Prototype (A8:C8)

In "Development of the Advanced Energy Design Guide for K-12 Schools – 50% Energy Savings," the primary school prototype based on DOE Commercial Reference Buildings was 73,930 ft2 with 650 students. The "Technical Feasibility Study for Zero Energy K-12 Schools" uses a primary school prototype that is 82,500 ft2 with 650 students. The Culver City Unified School District "Facilities Master Plan" shows four elementary schools of 47,825 ft2, 41,452 ft2, 45,077 ft2, and 46,045 ft2. In "Designing for Zero Carbon Volume 3 Case Studies of All-Electric K-12 Schools", the case studies included two elementary schools which were 49,040 ft2 and 33,638 ft2, and the middle school which was 24,298 ft2. Additional "noteworthy projects" included:

- 8891 ft2 K-3 school,
- 55,366 ft2 K-5 school,
- 13,515 K-5 school,
- 20454 ft2 grades 6-8 school,
- 11,600 ft2 K-8 school,
- 38,013 ft2 K-5 school
- 169,346 ft2 grades 6-8 school

Based on these, the 49979 ft2 K-8 prototype appears to be a reasonable size, but no justification was provided on how that area was chosen.

K8S18-Prototype (A12:C12)

The window-to-wall ratio (WWR) for the overall building of 7% seems way too small. It would be low for a classroom, and it seems low for the multipurpose room/administration building. I'm not sure where the original assumption comes from, but it seems counter to trends in classroom and administration (typically office) designs. The DOE prototype model had 35% WWR (Development of the Advanced Energy Design Guide for K-12 Schools – 50% Energy Savings, NREL/TP-5500-51437, February 2013), the DEER primary school prototype had 21% WWR (California K-12 and Community College Zero Net Energy Retrofit Readiness Study, NBI for CPUC, October 20217), WWR for buildings study vary from10% to 51% and are typically 20% (Actual Energy Performance of Secondary Schools Designed to Comply with ASHRAE Standard 90.1-2010, ASHRAE 1814-RP, January 2023). The importance of light, and especially natural light, is well established for health, learning, and behavior in classrooms, so a value as low as 7% needs further consideration.

K8S19-Prototype (A12:B12)

It would be good to state the window-to-wall ratio by categories of space types. Classrooms and administrative rooms would be of special interest, as would the rooms that seem to be highly glazed in the administration building.

K8S20-Prototype (A14:C16)

The floor-to-floor and floor-to-ceiling heights of 14 ft in the classroom buildings and 14 ft to 25 ft in the administrative/multipurpose room building may be reasonable but seem a bit high. From personal experience, older schools (before the 1960s) often have very high

ceilings, and I believe it is a trend for newer schools to have higher ceilings again, but there were many years where classrooms had fairly "office-like" ceilings of 9 to 11 feet. The California K-12 and Community College Zero Net Energy Retrofit Readiness Study, NBI for CPUC, October 20217 has the following table:

Table 19: Basic Assumptions of DEER Prototype Models

	Primary School	Secondary School		Relocatable Classroom	Community College
		Educational Building	Gymnasium		
Stories	1	2	1	1	3
Total Area (ft2)	50,000	127,500	22,500	1,920	300,000
Window-to-Wall Ratio (%)	21	20		4	26
Exterior Walls Construction Type	Wood frame	Heavy weight		Metal frame	Heavy weight
Roof Construction Type	Built-up flat roof	Built-up flat roof		Built-up flat roof	Built-up flat roof
Floor-to-Floor Height (ft2)	12	13	26	13	13

I have not found a good statistical basis for this value from the literature, so it would be good to justify the value selected.

The "Public Schools of North Carolina Facilities Guidelines" (2019) shows:

Ceiling Heights

Room Size	Ceiling Heights
850 SF or less	*9'-4"*
851 SF or more	*10'-0"*
Mobile Classrooms	*8'-0"*

K8S21-Prototype (n/a)

Major metrics for school building design are absent, such as

- number of students total and by grade
- number of classrooms total and by grade
- average students per classroom and average by grade

- students per administrator
- fraction of the area used by classrooms

It is difficult to assess the prototype without these fundamental metrics. Students are not separated from administrators or teachers, and only occupants are shown on the Zone tab. The number of classrooms represented is not made clear from the Zone tab either.

K8S22-Prototype (A7:C7)

While no specific justification for the shape was provided, the three classroom buildings and an administrative building with a multipurpose room surrounding a large courtyard area are not unreasonable for many climates in California that have schools without indoor corridors. This is probably not appropriate for all climate zones in California but common enough that it is reasonable for a single prototype.

K8S23-Prototype (A20:B23)

It is impossible to verify the total floor area, above-grade wall area, gross wall area, or roof area without dimensions for each of the buildings.

K8S24-Prototype (A10:C10)

While no specific source is used to justify the number of floors for the buildings, having a single story for the kindergarten classrooms and two stories for the classrooms in the north and west seem reasonable.

K8S25-Prototype (n/a)

More information was provided on other scorecards related to the existing prototypes and how this prototype is compared. It would be good to include that type of information on this tab.

K8S26-Zones (N1:T33)

The diagram included on the Zones tab does not show the locations of the actual rooms but only shows the outlines of rooms without labels. The diagram also does not show the

second floor separate from the first floor for those buildings with two floors. It makes the diagram not as useful as it could be in assessing the proximity of rooms to each other. Other scorecards had much better diagrams of the space layouts.

K8S27-Zones (H1:H41)

The volume column is not calculated correctly and seems to include numbers that do not correspond to the area of the rooms. The volume for 35 of the spaces is shown as 353 ft3 for rooms with areas as large as 6054 ft2, implying a floor-to-ceiling height of less than 1 inch for that room. The total volume is shown as 191,561 ft3, which implies a ceiling height of less than 4 ft.

K8S28-Zones (A41, C41, D41)

Three of the "total" row labels are incorrect.

- There are 13 space types, not 15
- There are 38 HVAC zones, not 5
- The "Total (5 HVAC Zones)" should probably say "Total AHUs" or "Total HVAC Systems"
- There are 3 schedule categories, not 4

K8S29-Zones (N1:T33)

The number of skylights shown in the diagram seems very small. It would not be uncommon in classrooms to have some diffused ceiling daylighting, either skylights or clerestory windows. In the multipurpose room, which is the only room with skylights, four small skylights are present, but that seems too small for the size of the room. It is hard to tell what fraction of that room ceiling has skylights, but it is commonly 3-7%.

K8S30-Zones (C1:C40)

While an air handler for every single room might be very desirable from a control perspective, it is unusual to see 38 air handlers in a building of 50,000. Presumably these air handlers are each single-zone heat pumps based on the HVAC types shown on the HVAC System tab. In the administrative/multipurpose room building, it is likely that some of the administrative spaces would share the same air handler. The small offices in the

classroom buildings are less than 300 ft2 and would probably share an air handler with an adjacent classroom. 27 of the rooms are under 1200 ft2, which would probably be served by 3-ton units or smaller, and of these 19 are actually under 600 ft2, which seems very small to be served by their own air handling units. One air handler serves only an 85 ft2 electrical room.

K8S31-Zones (A14:D14) and HVAC System

One room listed is a refrigerated storage room, which makes sense for a school facility since they serve many meals in the multipurpose room. It should have a different type of system and a different schedule than the rest of the multipurpose area.

K8S32-Zones (Rows 6,7, 25,26, 32, 37, 38)

There are seven "Electrical, Mechanical, Telephone Rooms" shown in the list of zones, which seems a little excessive for their impact on energy use. Three of them are shown as "IDF Rooms," which is a room with computer networking equipment. These rooms are also shown to contain 12 people in total, which seems highly unlikely.

K8S33-Zones (J2:J41)

The total number of people, 2137, seems very large for a building with only 49,979 ft2. This number is not the number of students but presumably includes students, faculty, administrators, and other staff.

The number of students is not specifically described even though it is a major metric for schools that would be part of any architectural planning. In the "Final Evaluation Report for The Alliance to Save Energy Green Schools Green Communities 2002-2003 School Programs," elementary school enrollments shown below range from 254 to 1019 students with an average of 649 students. For middle schools, the range of students is 449 to 1676 with an average of 1210 students.

Elementary Schools

Pennycooke	665
Federal Terrace	610
Valley Vista	396
Mary Collins	254
MP Brown	359

Majorie Tobias	302
Thomas Edison	420
John F. Kennedy	485
Yalupa	518
Bellevue	583
Kawana	633
Boyd	697
Curtis	1019
Dollahan	874
Hughbanks	820
Trapp	714
Morris	865
Calimesa	669
Belevedere	861
Emmerton	868
Mesquite Trails	698
Carmel	688
Eucalyptus	705
Hollyvale	465
Joshua Circle	806
Juniper	679
Lime Street	887
Maple	645
Minimum	254
Average	649
Maximum	1019

Junior High/Middle Schools

Petaluma	810
Frisbie	1530
Kolb	1477
Jehue	1676
Canyon	449
Parkview	1112
Serrano	1206
Ranchero	1304
Hesperia	1327

Minimum	449
Average	1210
Maximum	1676

In "California K-12 and Community College Zero Net Energy Retrofit Readiness Study" (2017), data from all California public schools showed that the average elementary school had 530 students and the average middle school had 763 students.

School Type	Number of Schools	Enrollment	Average Enrollment
Elementary	5,858	3,106,462	530
High	1,339	1,769,487	1321
Middle	1,298	990,159	763
Other	610	36,537	60
Continuation	452	55,899	124
K-12	262	152,693	583
Alternative	261	62,005	238
Community Day	193	3,669	19
Special Education	132	21,147	160
Junior High	48	28,679	597
Total	10,453	6,226,737	596

An older survey for the four major IOUs called "California Schools Market Characterization" (2005) showed the average number of students for elementary schools was 572 and middle schools was 964, with some differences by IOU.

Table 7-2. Number of Public Schools, Enrollment, and Average Enrollment, 2003-04: California and Four Major IOUs

	California*	PG&E	SCE	SDG&E	SoCalGas
Schools					
Elementary	5503	2567	1255	461	2453
Middle	1235	580	291	100	549
High	1059	529	235	85	417
Other**	1426	809	272	87	467
Total	9223	4485	2053	733	3886
Enrollment					
Elementary	3,146,620	1,204,308	796,010	281,968	1,667,065
Middle	1,190,867	432,182	298,790	110,702	645,862
High	1,700,913	652,972	438,348	158,467	883,016
Other**	260,374	106,077	65,751	18,629	118,409
Total	6,298,774	2,395,539	1,598,899	569,766	3,314,352
Average Enrollment					
Elementary	572	469	634	612	680
Middle	964	745	1,027	1,107	1,176
High	1,606	1,234	1,865	1,864	2,118
Other**	183	131	242	214	254
Total	683	534	779	777	853

* Number of schools is for California and is not equal to the sum of schools for each of the IOUs due to overlap of service districts and the fact that parts of California are not in the service districts of the four major IOUs

** Other includes Alternative, County Community, Community Day, Continuation, California Youth Auhority (CYA), Juvenile Hall, K-12, Opportunity, Special Educaiton, State Special.

Source: California Department of Education, 2005, Fact Book 2005, p. 8 for number of schools and p. 21 for enrollment. California Department of Education, CBEDS data file for 2003-2004 for IOU-specific information.

According to the Digest of Education Statistics produced by the National Center for Education Statistics, the average elementary and the average middle school have 15.1 pupils/teacher in 2022 and 7.5 staff members, including teachers per pupil. Based on this, for a school of 530 students, the total number of occupants would be 601 people.

A bottom-up approach for estimating the number of people in the school would be based first on the classroom area. For this prototype, there are 26,734 ft2 of "Classroom, Lecture, Training, Vocational Areas," which accounts for 53% of the school. This is very close to the 55% of the primary school buildings that are classroom/lecture shown in "California K-12

and Community College Zero Net Energy Retrofit Readiness Study" (2017). The average classroom size is 750-1000 ft2 based on "Architectural Graphic Standards 12th Edition Student Edition". The "Building Type Basics for Elementary and Secondary Schools" states, "California Grades 1 through 12 require not less than 960 NSF, or an equivalent space that provides not less than 30 sq ft per student." Based on this, the prototype should have 27 or fewer classrooms. Assuming 25 students per classroom, that is 675 students, and using the 7.5 staff per pupil statistics, that would be 765 total people.

Note that the schedules for almost the entire building are either the "classroom," which is from 75% to 90% during school hours, or the "MPR" schedule, which varies from 25% to 95% during school hours, so it is unlikely that movement between rooms could justify the number of total occupants shown in the scorecard, which are based on ventilation requirements. Since the occupancy is based on the ventilation requirements, it may be assumed that it is based on the most overcrowded condition possible since this would be the worst-case scenario. The number shown in the prototype is 2137, which is almost three times the bottom-up estimate of 765.

The number of people in certain non-classroom zones does not seem reasonable, such as 82 people in the lobby, 364 people in the office, 12 people in the electrical/mechanical rooms, or 29 people in storage rooms.

K8S34-Zones (A65:J80)

A breakdown by area, fraction of area, and number of people was added to the spreadsheet and may be helpful in review by others. It would be helpful to compare this with other schools. It would be helpful to understand the sources used in the decision-making process that arrived at this list of spaces and the areas of each. The values from the prototype are shown below:

	Area (ft2)	
Lounge, Breakroom, or Waiting Area	575	1%
Convention, Conference, Multipurpose and Meeting Area	7,187	14%
Corridor Area	1,162	2%
Electrical, Mechanical, Telephone Rooms	1,215	2%
Kitchen/Food Preparation Area	1,457	3%
Lobby, Main Entry	1,241	2%
Office Area (<250 square feet)	1,625	3%
Restrooms	3,359	7%
Storage (General)	2,910	6%
Storage, Commercial/Industrial (Refrigerated)	252	1%

Theater Area (Performance)	1,066	2%
Classroom, Lecture, Training, Vocational Areas	26,734	53%
Library (Reading Area)	1,197	2%
	49,979	

In "Development of the Advanced Energy Design Guide for K-12 Schools – 50% Energy Savings" (AEDG), the primary school prototype is shown in Table 3-5:

Space Type	Primary School	
	Area (ft^2)	Percent of Total
Auditorium	0	0%
Art classroom	1,744	2%
Cafeteria	3,391	5%
Classroom	35,457	48%
Corridor	12,078	16%
Gym/multipurpose room	3,843	5%
Kitchen	1,808	2%
Library/media center	4,295	6%
Lobby	1,841	2%
Mechanical room	2,713	4%
Office	4,747	6%
Restroom	2,045	3%
Total	73,962	100%

The percentage of classroom areas seems comparable when combining art and regular classrooms in the AEDG table. The corridor area is reasonable since the prototype has most of the circulation occurring outdoors. If the cafeteria and gym/multipurpose room in the AEDG are 10%, yet almost 14% is used in the prototype, which may be a bit high. Kitchen and lobby seem reasonable. The office in the prototype seems a bit small. Restroom may be a bit high, and library may be a bit low in the prototype.

K8S35-Zones (n/a)

The space for each type of room should be based on a bottom-up analysis starting with the number of students being served by the building. The following typical space programs from the North Carolina Department of Public Instruction Safe and Healthy Schools Division show some examples of this approach:

Elementary/Middle School PK-8										TYPICAL SPACE PROGRAM
No. of Elementary Students	150		150		250		300		350	
No. of Middle Students	150		300		370		470		600	
Total Number of Students	300		450		620		770		950	
Core Capacity	300		600		700		850		950	
Exceptional Children (S/C)	1		1		2		2		3	
Dance/Drama Classrooms									1	
Pre-Kindergarten Classrooms	1		1		2		3		3	
Auditorium (Y/N)	N		N		N		N		Y	
Number of Seats									380	
Classrooms:	No.	SF	No.	SF	No.	SF	No.	SF	No.	SF
Elementary (PK-5) Classroms										
Pre-K Classrooms @ 1200 SF	1	1,200	1	1,200	2	2,400	3	3,600	3	3,600
Kindergarten Classrooms @ 1200 SF	2	2,400	2	2,400	3	3,600	3	3,600	4	4,800
Grade 1 Classrooms @ 1000 SF	1	1,000	1	1,000	2	2,000	3	3,000	3	3,000
Grade 2-3 Classrooms @ 1000 SF	3	3,000	3	3,000	5	5,000	6	6,000	7	7,000
Grade 4-5 Classrooms @ 950 SF	2	1,900	2	1,900	3	2,850	4	3,800	4	3,800
Middle Grades (6-8) Classrooms										
Math/Science Classrooms @ 1000 SF		0		0		0	3	3,000		0
LA/SS/Math Classrooms @ 850 SF	6	5,100	9	7,650	12	10,200	12	10,200	18	15,300
Science Classrooms @ 1100 SF + Prep		0	3	3,300	3	3,300	3	3,300	6	6,600
Small Group/Exceptional										
Resource Rooms @ 450 SF	2	900	3	1,350	5	2,250		0	4	1,800
Exceptional S/C @ 1200 SF	1	1,200	1	1,200	1	1,200	2	2,400	2	2,400
Arts/Computer										
Art w/ 200 SF Storage	2	2,400	1	1,200	1	1,200	1	1,200	1	1,400
Music/Band w/ Storage									1	1,600
Music w/ Storage	1	1,500	1	1,200	1	1,200	1	1,200	1	1,000
Dance/Drama									1	1,800
Computer Classroom @ 1000 Sf										
Art/Project Room @ 1200 SF (Elem)	1		1	1,200	1	1,200	1	1,200	1	1,200
Vocational										
Keyboarding Lab									1	1,000
Exploratory Labs	1	1,400	1	1,400	1	1,400	2	2,800	2	2,800
Classroom Total Area		22,000		28,000		37,800		45,300		59,100
Media Center										
Media RLV (sf/stu & sf)	6	1,600	5	3,000	4	2,800	4	3,400	4	3,800
Support spaces		1,500		1,800		1,800		1,800		1,800
SubTotal		**3,100**		**4,800**		**4,600**		**5,200**		**5,600**
Food Service										
Dining (No. of Servings & SF)	2.3	1,800	3	2,800	3	3,267	3	3,967	3	4,433
Kitchen		1,261		1,518		1,938		2,208		2,208
Serving Lines (No. and SF)	1	400	2	620	2	620	2	620	2	620
Subtotal		**3,461**		**4,938**		**5,825**		**6,795**		**7,261**
Physical Education										
Gym (including Lockers)	1	6,500	1	6,500	1	6,500	1	8,500	1	10,000
Auxillary Gym										
Health/PE Classroom										
PE/Multipurpose (+ storage) (Elem)		4,650		4,650		4,650		4,900		4,900
Subtotal		**11,150**		**11,150**		**11,150**		**13,400**		**14,900**
Auditorium										
Seating (No. and SF)									380	3,420
Stage/Dressing/Storage										2,500
Lobby										1,000
Subtotal		**0**		**0**		**0**		**0**		**6,920**

Elementary/Middle School PK-8										TYPICAL SPACE PROGRAM
No. of Elementary Students	150		150		250		300		350	
No. of Middle Students	150		300		370		470		600	
Total Number of Students	300		450		620		770		950	
Core Capacity	300		600		700		850		950	

Administration, Miscellaneous										
Prinicpal		200		200		200		200		200
Assistant Principal				150		150	2	300	2	300
Reception		400		400		400		400		400
Secretary	1	150		150		150	2	300	2	300
Sims		150		150		150		150		150
Health		150		200		200		200		200
Office work room		290		225		238		256		269
Conference		200		200		200		250		250
Records		90		125		138		156		169
Office storage		90		125		138		156		169
Guidance room		300		300		300		300		300
Office/Testing @ 150 SF	1	150	1	150	2	300	2	300	3	450
Other student services		150		150		150		200		200
Team office/workrooms		600		900		1,350		1,350		1,350
Teacher lounge(s)		300		450		450		600		600
Itinerant Teacher Office (Elem)	1	100	1	100	1	100	2	200	2	200
Book Storage		650		800		850		925		975
General Storage/Receiving		750		900		950		1,025		1,075
Commons (Middle)				1,450		1,525		1,638		1,713
Resource Officer (Middle)	1	150		150		150		150		150
Sutotal		**4,870**		**7,275**		**8,089**		**9,056**		**9,420**

Total Net SF		39,711		56,163		67,464		79,751		103,201
Circulation, Toilets & Mech @ 35.0%		13,899		19,657		23,612		27,913		36,120
GRAND TOTAL SF		**53,610**		**75,820**		**91,076**		**107,664**		**139,321**
SF/Student (incl Pre-K and S/C)		**179**		**126**		**130**		**127**		**147**
Total Teaching Stations		**24**		**30**		**41**		**45**		**60**
Total Capacity (K-8 +Pre-K+S/C)		**312**		**478**		**661**		**799**		
Classrooms K-5		**9**		**9**		**15**		**19**		**21**
Classrooms 6-8		**6**		**12**		**15**		**18**		**24**
Other Teaching Stations		**9**		**9**		**11**		**8**		**15**

K8S36-Zones (K16, K17, K20, K21, K24, K30, K31, K35, K36)

The person density is shown as 50 for classrooms (K16, K17, K20, K21, K24, K30, K31, K35, K36), but it should be 25 according to the referenced table:

TABLE 120.1-A– Minimum Ventilation Rates

Occupancy Category – Educational Facilities	Minimum Occupant Load Density (persons / 1000 ft²)	Area-based Minimum Ventilation R_a (cfm/ft²)	Air Class	Notes
Daycare (through age 4)	14	0.15	2	--
Daycare sickroom	5	0.15	3	--
Classrooms (ages 5-8)	25	0.15	1	--
Classrooms (age 9 -18)	25	0.15	1	--
Lecture/postsecondary classroom	25	0.15	1	F
Lecture hall (fixed seats)	71	0.15	1	F
Art classroom	25	0.15	2	--
Science laboratories	25	0.15	2	--
University/college laboratories	25	0.15	2	--
Wood/metal shop	10	0.15	2	--
Computer lab	25	0.15	1	--
Media center	25	0.15	1	A
Music/theater/dance	33	0.15	1	F
Multiuse assembly	33	0.15	1	F

The person density is shown as 10 for corridors (K5), but it should be 5 according to the referenced table:

TABLE 120.1-A– Continued Minimum Ventilation Rates

Occupancy Category - *General*	Minimum Occupant Load Density (persons / 1000 ft²)	Area-based Minimum Ventilation R_a (cfm/ft²)	Air Class	Notes
Break rooms	33	0.15	1	F
Coffee Stations	33	0.15	1	F
Conference/meeting	33	0.15	1	F
Corridors	5	0.15	1	F
Occupiable storage rooms for liquids or gels	2	0.15	2	B

Likewise, for office zones, 10 person/1000 ft2 is used instead of 5:

TABLE 120.1-A– Continued Minimum Ventilation Rates

Occupancy Category - *Office Buildings*	Minimum Occupant Load Density (persons / 1000 ft²)	Area-based Minimum Ventilation R_a (cfm/ft²)	Air Class	Notes
Breakrooms	33	0.15	1	--
Main entry lobbies	33	0.15	1	F
Occupiable storage rooms for dry materials	2	0.15	1	--
Office space	5	0.15	1	F
Reception areas	5	0.15	1	F
Telephone/data entry	33	0.15	1	F

The use of double the number of people as specified in the tables continues for:

- Lounge, Breakroom, or Waiting Area Admin/MPR_Breakroom_Zn
- Convention, Conference, Multipurpose and Meeting Area
  Admin/MPR_Conference Room Zns 3_Zn
- Kitchen/Food Preparation Area Admin/MPR_Kitchen_Zn
- Lobby, Main Entry Admin/MPR_Lobby_Zn
- Convention, Conference, Multipurpose and Meeting Area
  Admin/MPR_Multipurpose Room Zns 3_Zn
- Storage, Commercial/Industrial (Refrigerated) Admin/MPR_Storage - Refrigerated_Zn
- Theater Area (Performance) Admin/MPR_Theater - Performance_Zn
- Library (Reading Area) Classrm W_Flr 1_Library - Reading Area_Zn

For storage zones which have 10 persons/1000 ft2, the closest is "Occupiable storage rooms for liquids or gels" which is 2 persons/1000 ft2:

- Storage (General) Admin/MPR_Storage - General_Zn
- Storage (General) Classrm W_Flr 1_Storage - General_Zn
- Storage (General) Classrm W_Flr 2_Storage - General_Zn
- Storage (General) Kindergarten_Storage - General_Zn

Restrooms do not have specific occupant density (I think they are often assumed to be people located in other places), so perhaps should be zero. The scorecard shows 10 persons/1000 ft2 for restrooms:

- Restrooms Admin/MPR_Restroom Zns 3_Zn
- Restrooms Classrm N_Flr 1_Restroom_Zn

- Restrooms Classrm N_Flr 2_Restroom_Zn
- Restrooms Classrm W_Flr 1_Restroom_Zn
- Restrooms Classrm W_Flr 2_Restroom_Zn
- Restrooms Kindergarten_Restroom_Zn

K8S37-Envelope (n/a)

While skylights are shown on the Zones tab, the Envelope tab shows no description for the skylight.

K8S38-Envelope (n/a)

The slab-on-grade floor (assumed) is not described, nor is any perimeter or under-slab insulation.

K8S39-Envelope (D3:D7)

While the use of metal frame construction with gypsum board may be reasonable for the classroom buildings, it is unlikely that the multipurpose room, which would serve as the gymnasium, would choose this construction. Gypsum board is not a durable enough surface for a room with tossed basketballs and volleyballs. In general, gymnasiums and multipurpose rooms that serve as gymnasiums generally have interior surfaces of concrete or concrete block.

K8S40-Envelope (D3:D7)

I believe that many older schools used brick, block, or concrete construction and that it is also becoming more common again given interest in resilience and having more durable construction. It would be good to have statistics to confirm the exterior wall assumption of using metal frame construction with gypsum board over a mass construction.

K8S41-Envelope (F7:X7)

The exterior wall U-values are incorrect for

- 1998-2007 vintage based on T24-2001
- 2008-2018 vintage based on T24-2013

- 2019+ vintage based on T24-2025

The construction type shown in the scorecard is "Metal frame + insulation + Gypsum board," but the values shown appear to be for "Walls - Wood-framed and Other"

K8S42-Envelope (n/a)

The roof solar reflectance and thermal emittance are requirements in Title 24-2013 and Title 24-2025 and should be included in the table for all vintages.

K8S43-Envelope (H20)

For the 1998-2007 vintage using Title 24-2001, the WWR (window-to-wall ratio) range used is 31%-40%, yet on the Prototype tab, the WWR is shown as 7% (see K8S18-Prototype). One of these is incorrect or needs further explanation.

K8S44-Envelope (J20)

For the 1998-2007 vintage using Title 24-2001, the WWR (window-to-wall ratio) range used is 31%-40%, and the SHGC should be 0.31, but 0.36 is shown in the cell. It is likely this was copied from the 21%-30% WWR row of the table.

K8S45-Envelope (H4, H8:H9)

The reference to Title24-1986 uses "Table 2-53V-16," but this is actually referencing a range of tables and should be shown as "Table 2-53V-1 to Table 2-53V-16" or perhaps "Table 2-53V-1 to 16".

K8S46-Envelope (H4)

The reference for Title24-1986 shows "Heat Capacity was considered 10.0-14.99 (BTU/F/ft^2)" but no justification for this value is provided, and it may be lower for a wall that is "Metal frame + insulation + Gypsum board."

K8S47-Envelope (M4:M5)

For vintage 1978-1997, Title 24-1986 is used, which shows R values that must be converted into U-values by using the reciprocal. For all climate zones, the 3-digit R-values are shown

as 4-digit U-values except CZ5 and CZ6, which show 5 digits. While technically not incorrect, a consistent number of digits should be used.

K8S48-Envelope (G9)

For the roof U-values for the Pre-1978 vintage, the selected Title24 is shown as "NA," but the values are the same as "1992" used in the 1978-1997 vintage. The cell should say "1992."

K8S49-Envelope (F8:X9)

The Pre-1978 and 1978-1997 vintages show the same roof U-values, and the values for 1978-1997 say they come from Title 24-1992, but none of the values shown seem to be based on that source. Instead, they appear to be based on Title 24-2001, which is used in the 1998-2007 vintage.

K8S50-Envelope (F3:X3)

For exterior walls, the u-values for the Pre-1978 vintage are shown as taken from "ASHRAE 90.1, 2022, Table A3.4.3.1, 3.5", R-0." And while the values shown are consistent with this source, the referenced table is for "Assembly U-Factors for Wood-Framed Walls," and the description in the scorecard is to use "Metal frame + insulation + Gypsum board." This is one of a number of comments related to the type of walls in the prototype being inconsistent with the type shown in the source.

K8S51-Envelope (I27:X31)

The scorecard does not contain enough geometric information to confirm the conversion of the six-sided infiltration rates to the four-sided infiltration rate needed to compute the "I-design" values. The geometry is sufficiently described in most other prototype scorecards to verify the calculations, but is not present for the K-8 school. The width, height, and length of each building are needed to compute the wall, roof, and floor areas that are then used to determine the ratio of the area of all building surfaces (wall, roof, and floor) to just the wall area. This information should be provided for each of the buildings.

K8S52-Envelope (I27:X31)

Four different I-design values for each vintage should be computed since the school consists of four different buildings with different widths, lengths, and heights. There should be values for each of the west classroom, north classroom, kindergarten, and MPR/Admin buildings since they have different shapes.

K8S53-Envelope (C13:C22)

The assumption of using fixed rather than operable windows is not substantiated and is inconsistent with my personal experience in schools, which either use operable windows in each classroom or a mixture of fixed and operable windows in each classroom. During shoulder seasons, it is common to use natural ventilation in schools, which is only possible with operable windows.

K8S54-Envelope (H8:H9)

The assumption of "Considering 30 years for roof/ insulation replacement" may not be appropriate for schools, which often face funding issues. It is likely a longer period between roof replacements, and it would be good to get statistics. In the following GAO report, 25 of 64 districts surveyed indicated that they needed roof replacements.
"K-12 EDUCATION School Districts Frequently Identified Multiple Building Systems Needing Updates or Replacement" June 2020, GAO-20-494. Also see K8S4-Notes.

K8S55-HVAC System (B3:B4) and Zones (A10:G10)

The use of ACM 2025 System 3b SZHP for most zones, probably for new construction, makes sense except for in the large Multipurpose room that appears to be used as a gymnasium and as a cafeteria. This zone "Admin/MPR_Multipurpose Room Zns 3_Zn" uses AHU-8 and is 5512 ft2. This would probably be served with System 7b SZVAVHP. This is based on assuming 400-600 ft2/ton; that would be a system of 9.2 to 13.7 tons (110 to 165 kBtu/h). Of course, that zone could be served by multiple units, so it is a judgment call. Given the likely large swings of occupancy, a variable speed fan might make sense. This logic would also apply to existing systems which are shown as System 3a SZAC with gas furnace and perhaps should be System 7a SZVAVAC for the multipurpose room.

K8S56-HVAC System (E15)

The new construction heat pump efficiency for cooling of EER 11 for 65-135 kBtu/h is consistent with table 110.2-B in Title 24-2025. It is not clear what the source is for the EER of 11.2 for <65 kBtu/h since that size unit is usually expressed as SEER2.

K8S57-HVAC System (D15)

The heating efficiency at 47 for 65-135 kBtu/h should be 3.40 COP, not 3.3 as shown in the table. Also, no values are provided for <65 kBtu/h for heating efficiency.

K8S58-HVAC System (E14)

The 2008-2028 vintage is based on Title 24-2013, which shows 11.2 EER for 65-135 kBtu/h, not 10.3.

TABLE 110.2-A ELECTRICALLY OPERATED UNITARY AIR CONDITIONERS AND CONDENSING UNITS – MINIMUM EFFICIENCY REQUIREMENTS

Equipment Type	Size Category	Efficiency [a]		Test Procedure[c]
		Before 1/1/2015	After 1/1/2015	
Air conditioners, air cooled both split system and single package	≥ 65,000 Btu/h and < 135,000 Btu/h	11.2 EER[b] 11.4 IEER[b]	Applicable minimum efficiency values as determined by Title 20 California Code of Regulations Section 1605.1	ANSI/AHRI 340/360
	≥ 135,000 Btu/h and < 240,000 Btu/h	11.0 EER[b] 11.2 IEER[b]		ANSI/AHRI 340/360
	≥ 240,000 Btu/h and < 760,000 Btu/h	10.0 EER[b] 10.1 IEER[b]		
	≥ 760,000 Btu/h	9.7 EER[b] 9.8 IEER[b]		

K8S59-HVAC System (F14)

The reference to "2013 T24 Tables 112-A, 112-B" should be to Table 110.2A (see previous comment).

K8S60-HVAC System (F20, F25)

The reference to "Fan Motor and Drive Efficiency [ACM 2025, Table 14]" should reference Table 13: Minimum Nominal Efficiency for Electric Motors (Percent), not Table 14.

K8S61-HVAC System (F21:F25)

The use of 0.9 for fan motor efficiency with the note "0.9 is an average fan efficiency according to ACM 2025, Table 14" is not appropriate if all the single zone systems are small. In that case, efficiency should probably be at the smallest end of this table, either 85.5% or 86.5%. It is unlikely that the motor size will be greater than 7.5 hp for the number of small units used in this building.

Table 13: Minimum Nominal Efficiency for Electric Motors (Percent)

Motor Horsepower	Efficiency (%)
1	85.5
1.5	86.5
2	86.5
3	89.5
5	89.5
7.5	91.7
10	91.7
15	92.4
20	93.0
25	93.6

K8S62-HVAC System (D21, D24)

The Total Static Pressure of 3.37 in w.c. for new construction and 2.5 or 4.09 in w.c. for existing seems kind of high for rooftop units with probably minimal ducting.

K8S63-HVAC System (n/a)

Economizers are common on single zone systems; are they being used in this prototype?

K8S64-Thermostat (n/a)

Please provide the temperature setpoint being used in the "Storage, Commercial/Industrial (Refrigerated) Admin/MPR_Storage - Refrigerated_Zn" zone.

K8S65-Thermostat (n/a)

The large storage room in the Admin/MPR building "Admin/MPR_Storage - General_Zn" is 1248 sqft and is shown to have its own AHU, which may imply that it is essentially a small warehouse. If this is the case, it might have thermostat setpoints that are less governed by comfort and may be more like the setback temperatures for all hours.

K8S66-Thermostat (n/a)

The temperature of a classroom can impact the learning environment in multiple ways. Too warm or too cold is likely to distract children. Due to this, the actual thermostat setpoints for schools are often different. The prototype uses 70F/75F for occupied hours and 60F/80F for unoccupied hours. In "ASHRAE Research Project Report 1814-RP Actual Energy Performance of Secondary Schools Designed to Comply with ASHRAE Standard 90.1-2010" (2023):

- School #3 uses 68F/74F for occupied and 60F/80F for unoccupied
- School #12/#17 uses 69F/75F for occupied and 55F/85F for unoccupied
- School #21 uses 70.5F/75.5F for occupied
- School #23 uses 72F/75F
- School #24 uses 68F/72F for occupied and 50F/80F for unoccupied

The ASHRAE Handbook on HVAC Applications (2023) chapter on Educational Facilities includes the following table:

Table 6 Temperature and Humidity Design Ranges for K-12 Schools

Category/ Humidity Criteria	Indoor Design Conditions		
	Temperature, °F		
	Winter	Summer	Comments
Classrooms, laboratories, libraries, auditoriums[a, e]			
30 to 60% rh[g]	71.5 to 77.9	75.0 to 77.0	
Administrative, offices[f]			
30 to 60% rh	71.5 to 77.0	75.0 to 79.0	
Gymnasiums			
30 to 60% rh	68.5 to 74.0	74.0 to 78.5	For gym with wooden floor, 35 to 50% humidity recommended at all times
Shops			
20 to 60% rh	68.5 to 74.0	74.0 to 78.5	
Cafeteria[b]			
20 to 30% (winter), 50% (summer) rh	70.0 to 73.5	78.5	
Kitchen[b]			
No humidity control	70.0 to 73.5	84.0 to 88.0	
Locker/shower rooms			
No humidity control	75.0		Usually not conditioned
Toilets			
No humidity control	72.0		Usually not conditioned
Storage			
No humidity control	64.0		
Mechanical rooms			
No humidity control	61.0		Usually not conditioned
Corridors			
No humidity control	68.0		Frequently not conditioned
Natatorium[c]			
50 to 60% rh	75.0 to 84.0	75.0 to 84.0	Based on recreational pool
Ice rink[d]			
35 to 45°F dp (maximum)	50.0 (minimum)	65.0 (maximum)	Minimum 10°F temperature difference between dew point and dry bulb to prevent fog and condensation

Notes:

[a]Calculations based on Tartarini et al. (2020), for students/children wearing typical western summer and winter clothing (0.4 and 0.75 clo, respectively) and activity of 1.2 met. Air speed assumed at 19.7 fpm and mean radiant temperature (MRT) assumed equal to air temperature. Temperature range is within acceptable ASHRAE *Standard* 55 range (–0.5 < Predicted mean vote < +0.5).

[b]Based on Chapter 3.

[c]Based on Chapter 5.

[d]Based on Harriman et al. (2001).

[e]For libraries, keep minimum humidity of 30°F dp and maximum of 55% rh.

[f]Calculations based on Tartarini et al. (2020) for adults wearing typical western summer and winter clothing (0.7 and 1.0 clo, respectively) and activity of 1.0 met. Air speed assumed at 19.7 fpm and MRT assumed equal to air temperature. Temperature range is within acceptable ASHRAE *Standard* 55 range (–0.5 < Predicted mean vote < +0.5).

[g]Considerations such as geographical location, building envelope, etc., should be evaluated and discussed with building owner to determine minimum design relative humidity.

And Building Type Basics for Elementary and Secondary Schools (2001) contains the following table:

RECOMMENDED DESIGN TEMPERATURES

Space[a]	Winter Design, °F	Summer Design, °F DB	Summer Design, RH = (%)
Laboratories	72	76–78[d]	50–55
Auditoriums, libraries, administrative areas, etc.	72	76–78	50–55
Classrooms			
Prekindergarten through grade 3	75	76–78	50–55
Grades 4 through 12	72	76–78	50–55
Shops	72	76–78[b]	50–55
Locker, shower rooms	75[d]	c,d	
Toilets	72	c,d	
Storage	65	c,d	
Mechanical rooms	60	c	
Corridors	68	78–80[b]	60[c]

[a]For spaces of high population density and where sensible heat factors are 0.75 or less, lower dry-bulb temperatures will result in generation of less latent heat, which may reduce the need for reheating and thus save energy. Therefore, optimum dry-bulb temperatures should be the subject of detailed design analysis. [b]Frequently not air-conditioned. [c]Usually not air-conditioned.
[d]Provide ventilation for odor control. DB = dry-bulb temperature. RH = relative humidity.

K8S67-Ventilation (E5:E8, E11, E13, E18, E22, E25, E26, E29, E32, E34, E37, E38, E40)

The 30 cfm/person is shown for the following zones, but the equation 120.1F shows 15 cfm/person:

- Corridor Area Admin/MPR_Corridor_Zn
- Electrical, Mechanical, Telephone Rooms Admin/MPR_EMT_Zn
- Electrical, Mechanical, Telephone Rooms Admin/MPR_IDF Room_Zn
- Office Area (<250 square feet) Admin/MPR_Office - LT 250 sf Zns 6_Zn
- Storage (General) Admin/MPR_Storage - General_Zn
- Office Area (<250 square feet) Classrm N_Flr 1_Office - LT 250 sf_Zn
- Office Area (<250 square feet) Classrm N_Flr 2_Office - LT 250 sf_Zn
- Electrical, Mechanical, Telephone Rooms Classrm W_Flr 1_EMT-01_Zn
- Electrical, Mechanical, Telephone Rooms Classrm W_Flr 1_EMT-02_Zn
- Storage (General) Classrm W_Flr 1_Storage - General_Zn
- Electrical, Mechanical, Telephone Rooms Classrm W_Flr 2_IDF Room_Zn
- Storage (General) Classrm W_Flr 2_Storage - General_Zn
- Electrical, Mechanical, Telephone Rooms Kindergarten_EMT_Zn

- Electrical, Mechanical, Telephone Rooms Kindergarten_IDF Room_Zn
- Storage (General) Kindergarten_Storage - General_Zn

One zone shows 50 cfm/person:

- Kitchen/Food Preparation Area Admin/MPR_Kitchen_Zn

K8S68-Ventilation (Column C)

The number of people is too high based on both the requirements of Title 24 and by other metrics (see K8S33-Zones).

K8S69-Ventilation (F14)

Freezer and refrigerated spaces do not have an occupant- or area-based ventilation requirement even though 0.15 cfm/ft is shown in the scorecard:

TABLE 120.1-A– Continued Minimum Ventilation Rates

Occupancy Category - *Miscellaneous Spaces*	Minimum Occupant Load Density (persons / 1000 ft²)	Area-based Minimum Ventilation R_a (cfm/ft²)	Air Class	Notes
Bank vaults/safe deposit	5	0.15	2	F
Banks or bank lobbies	5	0.15	1	F
Computer (not printing)	5	0.15	1	F
Freezer and refrigerated spaces (<50°F)	0	0	2	E

K8S70-Ventilation (E12:G12, E19:G19, E23:G23, E28:G28, E33:G33, E39:G39)

Restrooms are shown as 15 cfm/person and blank cfm/ft2 exhaust, but in Title 24 they do not have a cfm/person airflow requirement and only have an exhaust requirement:

Table 120.1-B – Minimum Exhaust Rates [ASHRAE 62.1: Table 6-2]

Occupancy Category	Exhaust Rate, cfm/unit	Exhaust Rate, cfm/ft²	Air Class	Notes
Toilets – private	25/50	-	2	E
Toilets – public	50/70	-	2	D

D – Rate is per water closet, urinal, or both. Provide the higher rate where periods of heavy use are expected to occur. The lower rate shall be permitted to be used otherwise.

E – Rate is for a toilet room intended to be occupied by one person at a time. For continuous systems operation during hours of use, the lower rate shall be permitted to be used. Otherwise the higher rate shall be used.

K8S71-Ventilation (G8)

The kitchen area does not show any exhaust requirement, but Table 120.1-B requires 0.70 cfm/ft2.

K8S72-Ventilation (n/a)

While not explicitly listed, it is likely that a K8 school would have some classroom area devoted to being an "art classroom," which requires a 0.70 cfm/ft2 exhaust rate according to Table 120.1-B. Locker rooms (probably present) have a 0.25 cfm/ft2 exhaust rate. It is also possible that science rooms in a school would qualify as a laboratory and require 1.00 cfm2/ft2.

K8S73-Ventilation (n/a)

The ASHRAE Handbook on HVAC Applications, Chapter 8 "Educational Facilities" devotes pages 8.12 to 8.19 to air cleaning and air monitoring for schools. While this may not indicate common practice, it indicates a trend that additional air treatment and monitoring is performed in school applications to reduce individual illness and the rapid spread of it in school applications.

K8S74-Ventilation (n/a)

No differentiation by vintage was included for outdoor airflow. Earlier vintages of schools probably had significantly different ventilation requirements. Since this often impacts the sizing of ducts, intake vents, and air-handling equipment, it is unlikely to be upgraded very often. This is the same comment as for other prototypes.

K8S75-Water Heater (B8:B44)

The baseline water heater type is shown as HPWH for all spaces, but some are shown in the Appendix 5.4A spreadsheet as "gas" and others as "electric"; no spaces are called

"HPWH". 17 of the spaces should be listed as "gas". This labeling is not consistent with other scorecards, and not listing both electric and gas is also not consistent.

K8S76-Water Heater (D8:D44)

Since the number of people is directly used to calculate the peak hot water usage and the number of people needs to be revised, the peak hot water use should also be revised.

K8S77-Water Heater (B50, B55)

The values shown in these cells, 353 gal/hr, should represent the total, but cell F46 shows only 315 gal/hr, which is a simple total of rows. Clarification should be provided on how these numbers were calculated.

K8S78-Water Heater (C50:C55)

The equation used to determine the heat output in the ACM on page 247 (pdf 263) is:

(x (design supply temp – 55) x 8.2877 pounds/gallon x 1 Btu/pound-

But the formula in the cell uses 8.345 instead of 8.2877.

K8S79-Water Heater (E50, G50:G55, I50:I55)

The cells for required energy input, standby loss, and off/on cycle cost coefficient to ambient contain formulas and values that could use additional explanation.

K8S80-Water Heater (B59)

The cold-water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater. This is the same comment as for other prototypes.

K8S81-Water Heater (n/a)

No locker room or associated showers are specifically mentioned as a space type, but the large multipurpose room is used as a gymnasium. While it is common for younger grades not to require a shower after gym classes, it is fairly common for older grades (6, 7, and 8) to have showers after gym class. The use of a large shower facility and the number of showers required, if included, would probably be responsible for the largest peak usage and set the sizing for the water heater in the Admin/MPR building.

K8S82-Water Heater (B51:B54)

The use of 75 gal/hr as the minimum value for the "Peak Hot Water Usage" is not justified and a little confusing. It is more likely that thresholds on the minimum storage volume would have been used rather than on the peak hot water usage. For the four water heaters, three are shown as 30 gallons and one at 51 gallons. Given the typical sizes for water heaters, it is likely that 40 or 50 gallons (50 is a more common commercial water heater size) would have been selected for the buildings with the smaller demands and 75 gallons for the building with the greater demands.

K8S83-Interior Lights (C4:C16, F4:F16)

While use of additional lighting power is not as common in a school application, it is still likely that they might be used in the lobby (decorative or display), corridors (decorative or display), multipurpose room (used as both gym and cafeteria), classroom (to illuminate whiteboards), and offices (decorative or display).

K8S84-Interior Lights (G4:G16)

Given the low available budgets in schools to do renovations, it seems very unlikely that existing buildings would have updated their interior lighting systems to Title 24-2016. This implies that K8 schools, on average, replace their interior lighting systems in 10 years. Based on K8S4-Notes, it seems more likely that an older version of Title 24 should be used to represent the existing building stock.

K8S85-Interior Lights (n/a)

Using Title-24 2016 for all existing buildings may not be as accurate as taking an average of multiple versions of Title 24 covering recent vintages, assuming that older buildings have replaced their lighting in the last 30 years (or some other time period). This is similar to comments on other scorecards.

K8S86-Interior Lights (n/a)

Daylighting is extremely common in school designs and has been for many decades. High windows, skylights, and clerestory windows are common strategies to add daylighting to classrooms and other spaces. This is based partially on an assumed correlation between good lighting, which includes natural lighting, and the attention and participation by students.

K8S87-Ext Lights (A11:C11)

The number of building entrances assumed is four and is too low for a K8 school with four buildings onto a courtyard/play area. Each classroom has an exterior entrance for three of the buildings, and the Admin/MPR building is likely to have several. Based on K8-Zones, 27 classrooms should have at least 27 doors, and the Admin/MPR building is likely to have at least three or four entrances. For evacuation purposes, there may be a requirement for

each classroom to have two exits.

K8S88-Ext Lights (A10:C10)

The parking assumption used is one space per 830 ft2 of gross building area, but one of the sources cited, the 2019 Outdoor Lighting CASE Report, shows 1 parking space per 360 ft2 of gross building area.

Table 38: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for General Hardscape – Table 140.7-A

Assumptions for Statewide Estimates - General Hardscape		
General Hardscape	**Assumptions**	**Area Multipliers for Construction S.F.**
for Large Office, Small Office, Food, Restaurant, College	1 parking space per 250 sf of gross building area	1
for Hotel, Retail, School, Other	1 parking space per 360 sf of gross building area	0.7
for NR Warehouse, Ref. Warehouse	1 parking space per 830 sf of gross building area	0.3

In addition, the parking lot size assumption of 400 ft2 per space is only applicable to cars and not any bus parking. For many K8 schools, they have 10-12 bus parking spaces (based on personal observation).

K8S89-Ext Lights (A12:C12)

The hardscape ornamental light assumption of 0.1 ft2 per ft2 of gross floor area should also be multiplied by 25% based on the cited Case report:

Table 36: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for Specific Applications – Table 140.7-B

Assumptions for Statewide Estimates - Specific Applications										
		Applied to % of Building S.F. in Category								
Lighting Allowance	**Assumptions**	Office, LG & SM	Retail	Restaurant	Food (Grocery)	Warehouse, Ref & NR	Hotel	School	College	Other
Hardscape Ornamental Lighting	**0.1 SF per SF of gross building area**	50%	50%	50%	25%		50%	25%	25%	5%

K8S90-Ext Lights (A12:C12)

The building facade light assumption of 30' height should also be multiplied by 0.25 based on the cited Case report, but the spreadsheet multiplies by 0.5 instead.

Table 36: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for Specific Applications – Table 140.7-B

Assumptions for Statewide Estimates - Specific Applications											
		Applied to % of Building S.F. in Category									
Lighting Allowance	Assumptions	Office, LG & SM	Retail	Restaurant	Food (Grocery)	Warehouse, Ref & NR	Hotel	School	College	Other
Building Facades	**30' building height, 2 floors per building** (20% of applicable facades are lit)	25%	50%	50%	25%		50%	25%	25%	5%

K8S91-Ext Lights (n/a)

One of the credits for schools that should appear on the list is for "Student Pickup/Drop-off zone which can be up to 25 ft by 250 ft at 0.056 W/ft2 (up to 350 W).

CONTINUED: TABLE 140.7-B ADDITIONAL LIGHTING POWER ALLOWANCE FOR SPECIFIC APPLICATIONS

All area and distance measurements in plan view unless otherwise noted.

PER APPLICATION: WATTAGE ALLOWANCE PER SPECIFIC AREA (W/ft²). Use as appropriate provided that none of the following specific applications shall be used for the same area.

Lighting Application	Lighting Zone 0	Lighting Zone 1	Lighting Zone 2	Lighting Zone 3	Lighting Zone 4
Student Pick-up/Drop-off zone. Allowance for the area of the student pick-up/drop-off zone, with or without canopy, for preschool through 12th grade school campuses. A student pick-up/drop off zone is a curbside, controlled traffic area on a school campus where students are picked-up and dropped off from vehicles. The allowed area shall be the smaller of the actual width or 25 feet, times the smaller of the actual length or 250 feet. Qualifying luminaires shall be within 2 mounting heights of the student pick-up/drop-off zone.	Not applicable	No Allowance	0.056 W/ft²	0.200 W/ft²	No Allowance

K8S92-Ext Lights (n/a)

It is not uncommon, even for K8 schools, to have outdoor fields for sports such as soccer, track and field, basketball, football, softball, or baseball, which often are lit for games and activities, which during winter, may go until after sunset.

K8S93-Equipment (E7, E9, E14)

The source for the "Kitchen/Food Preparation Area", "Office Area (<250 square feet)", and "Classroom, Lecture, Training, Vocational Areas" is shown as "Plug load and schedules project with CEC, 2025" but only the plug load power density is from that report. The gas power density and refrigeration power are still from T24 2022 - Appendix 5.4A.

K8S94-Equipment (B19)

One elevator is not consistent with two of the buildings being two floors (west classroom and north classroom buildings). Many existing schools with two stories would not have any elevator, but if elevators are provided, perhaps due to ADA requirements, then it would make sense to have one elevator in each of the two buildings. New schools with two stories are likely to have an elevator in each building due to ADA requirements.

K8S95-Equipment (C19, C24)

The cell labeled "Calculate Elevator load (kW)based on ACM 2022:" shows 60 kW, and the design load cell shows 60,000 W, but the ACM 2022 and ACM 2025 show 10kW per elevator. The formula in C24 is multiplied by 6 and is probably a copy error from another scorecard.

K8S96-Equipment (A17:D26)

Using the same elevator design load for new construction and existing vintages seems incorrect. Elevators, while serviced regularly, are not typically updated that often, and existing buildings could easily have elevators that are less efficient. A major upgrade in elevator requirements has happened in 90.1 in one of the last few cycles, and I expect the same to be occurring in Title 24. A similar comment appears for other scorecards.

K8S97-Equipment (F19)

The reference to

https://buildingandinteriors.com/find-the-magic-number-elevators-per-building/

is not valid since that article does not provide guidance on schools. In addition, that link is no longer valid. The link is now:

https://buildingandinteriors.com/number-of-elevators-required-in-a-building/

K8S98-Equipment (n/a)

No computer classroom has been specifically identified, but it is not an unusual space in a K-8 school and would have significantly different plug loads.

K8S99-Schedules (n/a)

The references to schedules "Updated based on the Plug Loads project" could not be verified since the document:

Proposed Updates to Equipment Power Densities and Schedules in the 2028 Energy Code Nonresidential and Multifamily Alternative Calculation Method (NRMFACM) Reference Manual

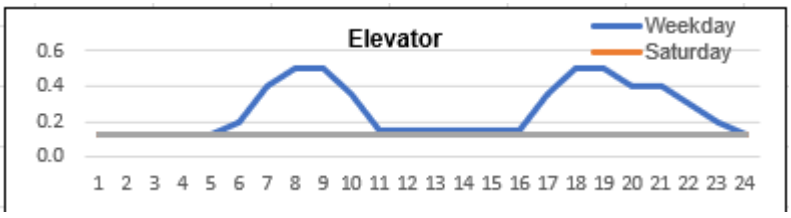
August 7, 2025

Specifically says for schools:

This effort is ongoing and will be documented in a separate report. After the occupancy schedules are finalized, new load schedules (e.g., lighting, receptacle, HVAC) will be developed to align with the new occupancy patterns.

K8S100-Schedules (B100:AC102)

The building elevator usage in the scorecard:



Is not consistent with the Title 24-20022 Appendix 5.4B values, which show only 30% from 9 am to 3 pm Monday through Friday and 15% for 4 pm and otherwise 0%.